Available online at www.sciencedirect.com

ScienceDirect

journal homepage: www.elsevier.com/locate/issn/15375110

Research Paper

Near-infrared spectroscopy as a tool for monitoring the spatial variability of sugarcane quality in the fields

Lucas P. Corrêdo [a,*], *Marcelo C.F. Wei* [a], *Marcos N. Ferraz* [a,b], *José P. Molin* [a]

[a] *Precision Agriculture Laboratory, Biosystems Engineering Department, 'Luiz de Queiroz' College of Agriculture, University of Sao Paulo, Brazil*

[b] *Smart Agri, Piracicaba, Sao Paulo, Brazil*[1]

ARTICLE INFO

Article history:
Received 3 June 2020
Received in revised form
2 April 2021
Accepted 5 April 2021
Published online 28 April 2021

Keywords:
Precision agriculture
NIR spectroscopy
Chemometrics
Saccharum officinarum
Portable spectrometer

It is known that Near-infrared spectroscopy (NIRS) is a reliable technique used in industrial laboratories to measure sugarcane quality. However, its use as a proximal sensing technology for monitoring the spatial variability of attributes in the fields has not yet been evaluated. The aim of this research was to examine the potential of NIRS for predicting and mapping Brix, Pol and Fibre content in a commercial sugarcane field. The quality attributes models were adjusted considering the spectral reflectance from the 1100–1800 nm wavelengths by using partial least squares regressions (PLSR). A total of 350 samples were collected in a sugar mill laboratory for calibration and cross-validation models development. For the external validation, 91 georeferenced samples were obtained from a commercial field. The results indicated that the developed models are capable of predicting Brix and Pol, with a coefficient of determination (R^2_P) of 0.71 for both parameters, and with a root mean square error of prediction ($RMSE_P$) of 0.80% and 0.58%, respectively. In contrast, the results for Fibre were unsatisfactory (R^2_P of 0.24 and $RMSE_P$ of 1.15%). Predicted values showed spatial dependence of the sugarcane quality attributes. Predicted and observed values of Brix and Pol presented a coefficient of correlation of 0.85. Results showed that NIRS has potential to be applied as a proximal sensing method supporting crop management based on the spatial variability of the quality attributes.

© 2021 IAgrE. Published by Elsevier Ltd. All rights reserved.

1. Introduction

In-field yield monitoring of sugarcane fields represents the main approach for managing the spatial and temporal variability of the crop (Maldaner & Molin, 2020). However, Brazilian sugarcane producers rarely use yield monitors (Sanches, Magalhães, & Franco, 2019) despite the existence of them. In addition to the importance of yield, quality analysis is also important for

* *Corresponding author*. Biosystem Engineering Department, Av. Padua Dias, 11, C.P 09, Piracicaba, SP, 13418-900, Brazil.
E-mail addresses: lucaspcorredo@usp.br (L.P. Corrêdo), marcelochan@usp.br (M.C.F. Wei), marcos@smart.agr.br (M.N. Ferraz), jpmolin@usp.br (J.P. Molin).

[1] Present address

https://doi.org/10.1016/j.biosystemseng.2021.04.001
1537-5110/© 2021 IAgrE. Published by Elsevier Ltd. All rights reserved.

Nomenclature

A	Range (m)
Brix	Soluble solids content (%)
C	Partial sill (dimensionless)
C0	Nugget (dimensionless)
CV	Coefficient of variation (%)
g	grams
h	Number of latent variables (dimensionless)
ha	hectare
j	Denotes an specific wavelength (nm)
kgf cm^{-2}	kilogram-force per centimetre square
LAI	Saccharimetric reading (°Z)
LV	Latent variables
m^2	square metre
Max	Maximum
Mg	Megagram
Min	Minimum
mL	Milliliter
N	Number of predicted samples
NIRS	Near-infrared spectroscopy
nm	Nanometre
p	Number of wavelengths (dimensionless)
PC	Pol of cane (%)
PCA	Principal component analysis
PLSR	Partial least squares regression
Pol	Apparent sucrose in the juice (%)
R^2	Coefficient of determination (dimensionless)
R^2_p	Coefficient of determination of prediction (dimensionless)
RMSE	Root mean square error (%)
$RMSE_p$	Root mean square error of prediction (%)
$RMSE_{cv}$	Root mean square error of cross-validation (%)
RS	Reducing sugars
s	seconds
SD	Standard deviation (%)
SDI	Spatial dependence index (%)
SNV	Standard normal variation
TRS	Total recoverable sugar (kg Mg^{-1})
VIP	Variable importance for the projection (dimensionless)
w	Loading weight (dimensionless)
y_i	Measured value of the ith sample (%)
$\hat{y}_i$	Predicted value of the ith sample (%)
Z	Fraction of variance in the prediction explained by the latent variable (dimensionless)
°C	Celsius degree

sugarcane crops since their combination is used to calculate the payment to sugarcane growers (Nawi, Chen, & Jensen, 2014).

Pricing and trading between growers and sugar mills is performed as a function of the total recoverable sugar (TRS) produced (Phetpan, Udompetaikul, & Sirisomboon, 2018) and is determined by the average result of a sampling obtained at the sugar mill. The TRS is calculated by attributes related to the sugarcane composition which should meet the quality standards of the industrial process. These attributes are the percentage of water-insoluble matter of cane and bagasse (Fibre), soluble solids content (Brix), apparent sucrose in the juice, measured by the optical activity which causes polarised light to be rotated (Pol). Purity, Pol of cane (PC) and reducing sugars (RS), are quality attributes calculated based on the former parameters (Brix, Pol and Fibre) by empirical equations (Rodrigues, Magalhães, & Franco, 2013). Then, the TRS value is obtained as a function of the estimated PC and RS of cane. In Brazil, these attributes are determined by conventional laboratory analysis described by the National Council of Sugarcane Producers of São Paulo State (CONSECANA, 2015), which are in agreement with the international rules from the International Commission for Uniform Methods of Sugar Analysis (ICUMSA).

The analysis to evaluate sugarcane quality at the field level aiming to manage its variability must be different from the conventional laboratory routine. The conventional analysis includes high-cost, time-consuming and environmental unfriendly methods (Corrêdo, Canata, Maldaner, de Lima, & Molin, 2020). Therefore, the management of spatial variability of sugarcane quality to assist in decision making must be from a feasible method to obtain quality results of the raw material, i.e. development of a proximal sensing method that rapidly measures product quality with high accuracy, requires minimal sample preparation and is environmentally friendly (Sanseechan et al., 2018).

Near-infrared spectroscopy (NIRS) has been used to determine the quality of many agricultural products (Cortés, Blasco, Aleixos, Cubero, & Talens, 2019), due the possibility in detecting signals from most of the major structures and functional groups of organic compounds (Pasquini, 2018). The interaction between electromagnetic radiation corresponding to NIRS, in the wavelength range from 750 nm to 2500 nm, and the sample can be reflected, transmitted or absorbed (Osborne, 2006). As a result of the interaction, the NIR spectrum can be obtained and used to extract information about the chemical composition of the sample (Wang, Peng, Xie, Bao, & He, 2015). Then, chemometric methods are applied to interpret the signal, relating them with other attributes of interest and to extract the responses (Pasquini, 2018).

Several studies have been carried out using NIRS to predict sugarcane quality attributes (Maraphum et al., 2018; Nawi, Chen, & Jensen, 2013; Nawi, Chen, Jensen, & Mehdizadeh, 2013; Phuphaphud, Saengprachatanarug, Posom, Maraphum, & Taira, 2019; Sorol, Arancibia, Bortolato, & Olivieri, 2010; Taira, Ueno, & Kawamitsu, 2010; Valderrama, Braga, & Poppi, 2007), presenting promising indicative results. Phetpan et al. (2018) used an approach to measure quality attributes by NIRS directly in the elevator conveyor of a sugarcane harvester. For this purpose, they developed a sugarcane harvester simulated elevator to measure the content of Brix in motion on the elevator conveyor resulting in a coefficient of determination of prediction (R^2_P) of 0.785 and root mean square error of prediction ($RMSE_P$) of 0.30.

Although several results indicate the NIRS as the most viable technology for predicting the quality of sugarcane, including its potential as a PA tool, there are lack of reports available analysing georeferenced samples obtained in the field to evaluate the viability of using NIRS to monitor the spatial variability of sugarcane quality attributes. Thus, the

objective of this study was to evaluate the potential of NIRS to predict sugarcane quality attributes (Fibre, Brix and Pol) and to characterise their spatial variability in a commercial field.

2. Material and methods

This study was conducted in two stages. The first refers to the development of calibration models to predict sugarcane quality attributes. The second stage is the application of the fitted model to georeferenced sampled data obtained in a commercial field to evaluate its spatial variability by means of NIRS.

2.1. *Laboratory sampling and conventional analysis*

Calibration data set was composed of samples obtained from a sugar mill quality laboratory located in Pradópolis, SP, Brazil, in July of 2018. The conventional analysis protocol used in this research followed the one used by the sugar mill laboratory quality standard analysis, as proposed by the National Council of Sugar Cane Producers of São Paulo State (CONSECANA, 2015). These protocols are in agreement with the international rules from International Commission for Uniform Methods of Sugar Analysis.

Conventional sugarcane quality analysis relies on a sampling procedure at the sugar mill. From each truck carrying harvested sugarcane billets, a sample of about 10 kg was extracted randomly from a truckload on the arrival of the harvested cargo at the sugar mill through an oblique probe and labelled (Figs. 1a and b). The samples were milled in a mechanical knife crusher and homogenised in a concrete mixer. Then, 500 g and 50 g samples of the homogenised mixture of defibrated cane were collected to proceed with the conventional and NIRS analysis, respectively.

For the conventional sugarcane attributes analysis, each 500 g sample was pressed in a hydraulic press under constant pressure, at 24.5 MPa for 60 s to obtain the raw juice. An automatic refractometer (RX-5000α, ATAGO Co Ltd, Tokyo, Japan), with maximum resolution of 0.1 °Brix, was used to determine the soluble solids content in the raw juice for each sample. After that, 14 g of a mixture composed of 4:2:1 proportion of Celite (mineral filtering agent), aluminium chloride and calcium hydroxide, respectively, was added to 200 mL of raw juice and homogenised through agitation process by a magnetic stirrer until the solution was well-mixed. Then, the solution was filtered through filter paper to obtain a transparent solution (Fig. 1c). A volume of 70 mL of this filtered solution was used for saccharimetric reading, measured by a digital polarimeter (Schmidt + Haensch, Polartronic NHZ 8, Berlin, Germany). The result was obtained as percentage of apparent sucrose in the juice (Pol). The remaining defibrated cane from the press step (without juice), named wet bagasse, was transferred to metal baskets, with holes at the base. Then, the wet bagasse was weighed on a semi-analytical balance to determine the weight of the wet bagasse (B_{wet}). The balance was previously tared to discount the mass of the baskets. Then, samples were held in a forced air circulation drying equipment at a temperature of 105 °C, until constant weight. The dryer samples were weighed, and the Fibre content (insoluble solids) was determined.

2.2. *NIR measurements*

Spectral absorbance was measured with a portable spectrometer AgriNIR (Dynamica Generale S.p.A., Poggio Rusco, Mantova,

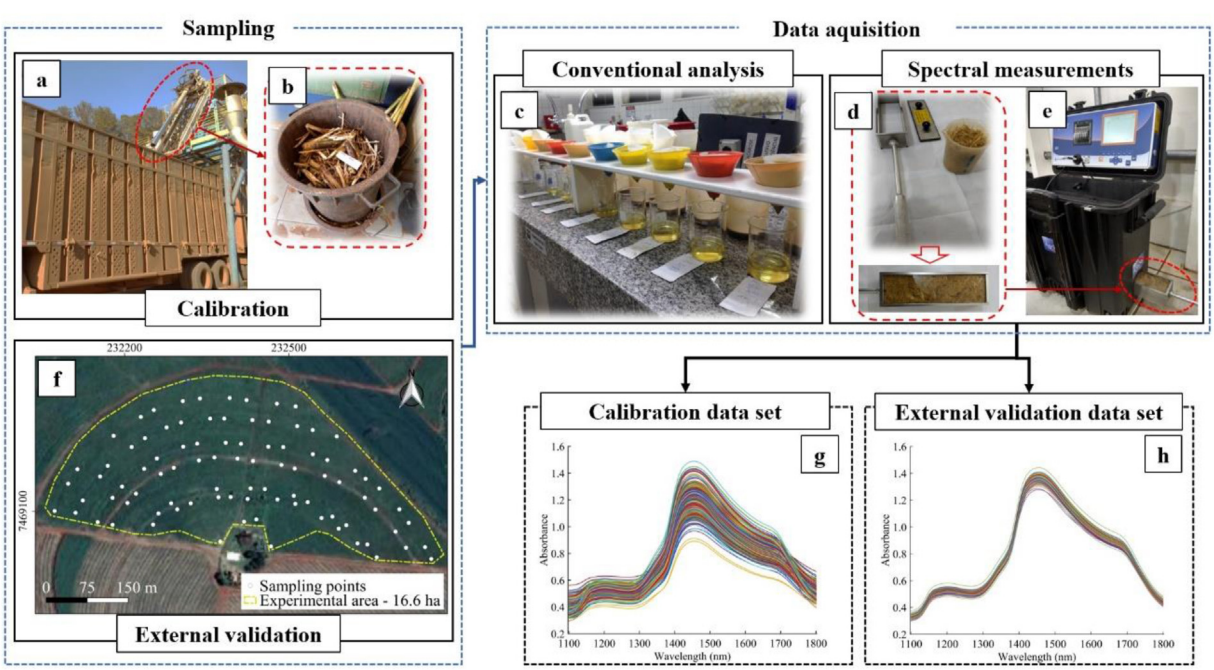

Fig. 1 – Process of sample obtaining to calibration (in the sugarcane quality laboratory) by an oblique probe (a) pulling out sample in the cargo truck (b); conventional laboratory analysis (c) after sample prepares; preparation of defibrated sample (d) to spectral measurements by NIRS (e); sampling points in the field to external validation (f); spectral data set of calibration samples (g) and external validation samples (h).

Italy), mostly used for forage analysis, operating in the spectral range of 1100–1800 nm with spectral resolution of 10 nm.

The 50 g samples of defibrated cane were transferred to a sample holder (Fig. 1d). The sample holder has a metal plate which compressed the sample on a glass surface reducing the number of pores and standardising the surface of the matrix to be scanned. Then, the sample holder was inserted in the dark chamber of the equipment (Fig. 1e) with a halogen light source inside for spectral measurements of the sample surface through the glass surface. The equipment results are composed of an average of three spectral readings. In addition, the procedure was repeated five times for each sample.

All spectral readings and conventional measurements were conducted at the same time at the sugarcane quality laboratory under controlled temperature (20 °C ± 5 °C). Thus, sugar degradation through microbial activities and molecular vibrations caused by temperature fluctuations during measurements were considered negligible.

The sample characteristic spectrum was composed of the average of readings to be associated with the result obtained from the conventional analysis. Hence, 350 samples of different cultivars of sugarcane were obtained for the calibration step.

2.3. Study field and external validation data set

The validation data set used in this study was obtained from the spatial-analysis database of the Precision Agriculture Laboratory at the University of Sao Paulo. This data set was composed of in-field database obtained from 16.6 ha of cultivar RB855453 harvested in the 2015 season. This cultivar is characterised as having high yield and sucrose content, and medium Fibre content (RIDESA, 2015). The georeferenced samples were randomly collected at the edges of each harvested line (Fig. 1f). A total of 91 samples were collected in the sugarcane field (approximately 5.5 samples ha^{-1}) composed of four stalks for each sampling. The whole stalks were manually cut at ground level and the growing point, and the leaves were removed. In-field samples were taken to the laboratory and milled to break the Fibre, homogenised and pressed, to replicate the sugar mill conventional laboratory procedure. All these samples were submitted to the conventional and NIRS analysis in the same day. Also, both procedures were performed in the same way as for samples in calibration step, as describe in topics 2.1 and 2.2.

2.4. Chemometrics

The multivariate analysis techniques applied to chemical compounds prediction are called chemometrics, and are widely used in spectroscopic data analysis (Mehmood & Ahmed, 2016). Generally, this procedure involves spectral pre-treatments, construction of calibration models and regression methods, and prediction of interest compounds by unknown samples (Cortés et al., 2019; Wang et al., 2015).

Initially, spectral data was used to build a matrix of independent variables (**X**) where the lines are composed of the 350 mean spectra of samples (Fig. 1g) and the columns were composed of 71 wavelengths. Thus, the vectors of dependent variables (**y**) were obtained from each sugarcane quality attribute determined by the conventional analytical methods used at the sugar mill laboratory. The external validation data set (Fig. 1h) was composed of the data obtained in-field.

Different pre-processing techniques were applied to the spectral data before calibrations. The main objective of this step is to transform data into useful information to develop models with reliable performance to the subsequent multivariate analysis, improving the model accuracy (Cortés et al., 2019; Phetpan et al., 2018), and ensuring non-distortion of the chemical information (Oliveri, Malegori, Simonetti, & Casale, 2019). The pre-processing adjustment was evaluated by the lowest value of the root mean square error (RMSE) of cross-validation (Eq. (1)).

$$\mathrm{RMSE} = \sqrt{\frac{\sum_i^N \left(y_i - \hat{y}_i\right)^2}{N}} \tag{1}$$

where y_i and $\hat{y}_i$ are measured and predicted values for each i sample, respectively. When internal cross-validation is used, N represents the number of samples in cross-validation set. Then, root mean square error is named as $RMSE_{CV}$. For external validation, with prediction samples set, N represents the number of predicted samples, and, in this case, the statistical parameters are named root mean square error of prediction ($RMSE_P$).

A preliminary data overview through principal component analysis (PCA) was performed for both data set (calibration and validation), for examination of pre-processed spectral data to find the similarities, differences and outliers among all the samples. The outliers were removed from the data set according the Hotelling's T^2 and Q-residuals tests, at a significance level of 5% (ASTM, 2017). Then, the modelling was performed applying partial least squares regression (PLSR).

PLSR is one of the most applied multivariate regression methods used to predict food quality attributes. It is a chemometrics covariance-based method that seeks to optimise both the variance explained and correlation with the dependent variables (Wold, Sjöström, & Eriksson, 2001). The spectral data are decomposed in latent variables (LV), which explain most of the variance of the raw data, modelling the linear relationship between the independent variables (matrix **X**) and dependent variables (vector **y**) using the scores of the most correlated features (Morellos et al., 2016; Phetpan et al., 2018). PLSR models determine new variables, which are estimates of the latent variables, named **X**-scores (predictors of the response ones). In addition, the deviations between measured and predicted values (**y**-residuals) are also obtained (Eq. (2)) (Steidle Neto, Lopes, Pinto, & Zolnier, 2017).

$$\mathbf{y} = \mathbf{bX} + \mathbf{e} \tag{2}$$

where **y** is the vector of dependent (response) variables (%), **b** indicates a vector of regression coefficients (dimensionless), **X** is the matrix of spectral variables of each sample (independent variables, %), and **e** is a vector that indicates the y-residuals. The model was performed by venetian blinds cross-validation, with 10 splits. The optimal number of LV selected

was the value that minimised $RMSE_{CV}$ (Eq. 1) the most. Then, the calibrated model was used to predict the quality attributes of the samples collected in-field.

To perform the spatial variability management of interest attributes, it is essential that differences from predicted and observed attribute present low error and the model presents a satisfactory performance. Thus, the regression models were evaluated by the RMSE of prediction (Eq. (1)) and coefficient of determination of prediction (Eq. (3)).

$$R^2 = 1 - \frac{\sum_i^N \left(y_i - \hat{y}_i\right)^2}{\sum_i^N \left(y_i - \bar{y}_i\right)^2} \tag{3}$$

where y_i and $\hat{y}_i$ are measured and predicted values for each i sample, respectively. The $\bar{y}_i$ is the mean y values. For external validation, with prediction samples set, N represents the number of predicted samples, and, in this case, the statistical parameter is named coefficient of determination of prediction (R^2_P).

The variable importance for the projection (VIP) was calculated to verify the wavelengths with greater impact on the external validate models (Eq. (4)).

$$VIP_j = \sqrt{p\sum_{k=1}^{h}\left[Z\left(\frac{w_{jk}}{||w_k||}\right)^2\right]\cdot\left(\sum_{k=1}^{h} Z\right)^{-1}} \tag{4}$$

where VIP is the variable importance for projection (dimensionless), j is a specific wavelength (nm), p is the number of wavelengths (dimensionless), h is the number of latent variables (dimensionless), Z is the fraction of variance in the prediction explained by the latent variable (dimensionless), and w is the loading weight (dimensionless). This parameter was analysed together with the regression vector generated by the modelling.

In this study, Brix, Pol and Fibre models were developed using PLSR, which was performed in MATLAB framework (MATLAB R2015a, The MathWorks Inc., Natick, USA) and PLS-Toolbox 8.8 (Eigenvector Research Inc., Wenatchee, WA, USA).

2.5. *Spatial variability analysis*

Geostatistical analyses were performed on the georeferenced database containing observed and predicted values from the laboratory analysis and by means of the best models from the chemometrics analysis, respectively. Semi-variograms were fitted for each set of observed and predicted sugarcane qualitative parameter using VESPER software (Minasny, Mcbratney, & Whelan, 2005) to estimate the structure of the spatial variation. The semi-variogram models that resulted in the lowest RMSE were selected. Spatial dependence was classified according to Cambardella et al. (1994). Maps were produced by interpolating the data using ordinary kriging at a spatial resolution of 1 m^2. Then, sugarcane quality attributes were mapping using the Quantum GIS Software 2.18 version (QGIS Development Team, 2016).

The similarity between measured and predicted maps for the sugarcane quality attributes was evaluated by visual comparison (Tümsavaş, Tekin, Ulusoy, & Mouazen, 2019) and using the coefficient of correlation (Colaço, Molin, Rosell-Polo, & Escolà, 2019).

3. Results and discussion

3.1. *Overview of the statistic characteristics and spectral data*

Results of sugarcane quality attributes obtained by conventional analysis for calibration step and field samples (external validation step) are shown in Table 1. The in-field values found for Brix and Pol ranged between the maximum and minimum values obtained for calibration samples. For Fibre, the minimum value observed in the external validation data set were lower than the minimum observed in the calibration samples.

After the pre-processing and applying PCA, only one sample in the external validation data set was identified by Hotelling T^2 and Q-residual statistic tests as an outlier, indicated as sample poorly modelled with high influence on the model, thus it was removed from the data set.

The coefficient of variation (CV) presented in the calibration data set was higher than those in the validation data set independently of the attribute, characterizing the diversity induced by samples from different cultivars obtained in the calibration data set. It is noteworthy that at this stage, it was desirable to obtain samples from different cultivars to obtain greater diversity of quality attributes to build a robust prediction model (Corrêdo, Maldaner, Bazame, & Molin, 2021; Phetpan et al., 2018).

The mean spectra, as well the standard deviation, for calibration and external validation samples are shown in Figs. 2a and b, respectively.

Looking at the quality attributes, the PLSR model was more accurate when it was pre-processed with the second derivative using standard normalise variation method (SNV) followed by Savitzky–Golay filter (Savitzky & Golay, 1964), with a window size of nine spectral bands and a polynomial degree of two. The SNV removes the path length variations on which the average value of a spectrum is subtracted from the original spectrum, and then the result is divided by the standard deviation (Wang et al., 2015), and second derivative removes the offset and slope on the spectral data.

Table 1 – Descriptive statistics of the samples for the calibration and prediction sets, after outlier removal.

Sample Set	Calibration			External validation		
Quality attribute	Brix	Pol	Fibre	Brix	Pol	Fibre
N	350			90		
Max (%)	23.8	20.9	19.1	19.2	18.0	11.9
Mean (%)	20.46	17.70	12.77	17.48	15.58	10.70
Min (%)	13.8	10.1	10.2	15.7	13.4	9.7
Median (%)	20.73	17.98	12.60	17.60	15.53	10.70
SD (%)	1.4746	1.6118	1.3153	0.69	0.98	0.5658
CV (%)	7.22	9.12	10.30	3.99	6.30	5.29

N: number of samples; Max: maximum; Min: minimum; SD: standard deviation; CV: coefficient of variation.

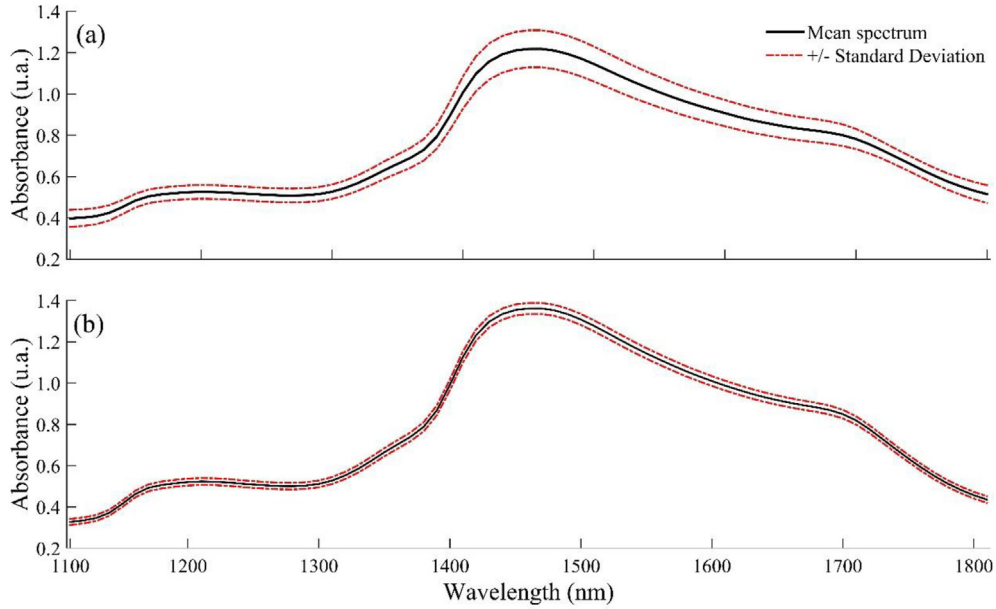

Fig. 2 – Mean (black line) spectra, with standard deviation above and below (dashed red lines), of spectral data of defibrated sugarcane samples of calibration (a) and external validation (b). u.a.: arbitrary units. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

3.2. *Performance of the calibration and prediction models*

A proper model should have a low RMSE and a high R^2, for both calibration and prediction steps (Nawi, Chen, & Jensen, 2013). The results obtained for the models predicting Brix and Pol were satisfactory (Fig. 3a and b), presenting $RMSE_P$ of 0.80 and 0.58, respectively, and R^2_P of 0.71 for both models (Table 2). Using absorbance spectra, Nawi, Chen, and Jensen (2013) predicted Brix content from stalk samples achieving $RMSE_P$ of 1.51%, value higher than the one found in this study ($RMSE_P$ of 0.80%). In a study predicting Pol, also from the whole stalk samples, Maraphum et al. (2018) evaluated the effect of skin wax on its estimation using NIRS. As a result, they found values of $RMSE_P$ ranging from 1.2% to 1.5% by scanning stalks without wax. These values were higher than the value of $RMSE_P$ (0.58%) found in this study from defibrated samples. The use of defibrated samples in NIRS analysis present several benefits such: elimination of the negative effect that skin compounds, i.e. wax, causes to the reading; and promotion of a more homogenous stem sample, since the plant present uneven sucrose concentration along the stem, with accumulation in an upward direction (Nawi, Chen, & Jensen, 2013; Wang, Nayak, Koch, & Ming, 2013). On the other hand, the Fibre predictive model presented an unsatisfactory performance, compared to the other attributes, with values of R^2 below of 0.3 and RMSE above of 1% (Fig. 3c).

Phuphaphud et al. (2019) obtained better results for predicting sugarcane Fibre content than those found in this study (Table 2), with $RMSE_P$ and R^2_P values of 0.81 and 0.63, respectively. However, the authors used the stalk measurement approach, divided into three sections, a reality that is difficult

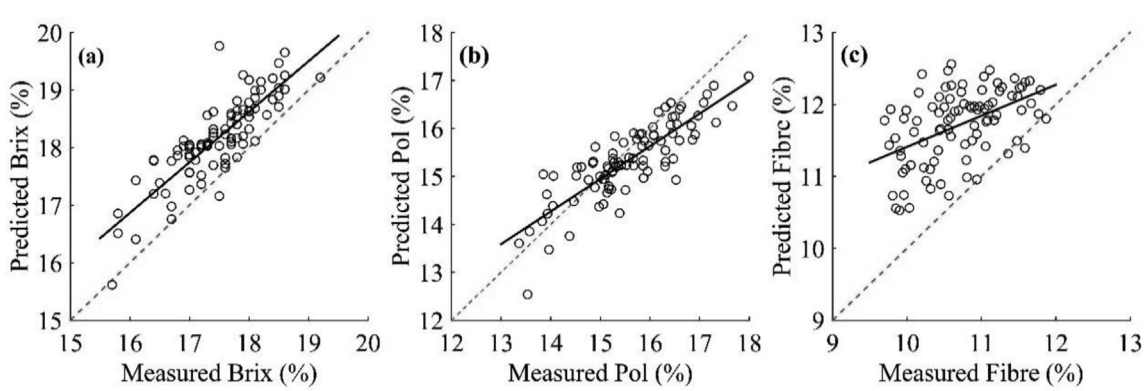

Fig. 3 – Scatter plots of measured versus predicted solids soluble content (Brix) (a), sucrose (Pol) (b) and Fibre content (c) of external validation data set.

Table 2 – Model results for partial least squares regression (PLSR) in cross-validation (cv) and external validation (p) sets for the prediction of solids soluble content (Brix), sucrose (Pol) and Fibre content.

Quality attribute	LV	Cross-validation		External Validation	
		$RMSE_{CV}$ (%)	R^2_{CV}	$RMSE_P$ (%)	R^2_P
Brix	7	0.66	0.80	0.80	0.71
Pol	7	0.80	0.75	0.58	0.71
Fibre	8	1.13	0.26	1.15	0.24

LV: latent variables; RMSE: root mean square error; R^2: coefficient of determination; cv = cross-validation; p = prediction.

to implement for an embedded sensor on a sugarcane harvester due to its layout and mechanical characteristics not allowing to distinguish each section after passing through the chipper roll. Therefore, the viability of an embedded NIRS sensor depends on ways of obtaining homogeneous and representative samples of each portion harvested on the crop (Corrêdo et al., 2020).

As an overtone spectroscopy, NIRS incorporate abundant information concerning OH, CH and NH vibration absorptions (Wang et al., 2015). In the modelling by PLSR, the observed values are approximated by the values of the predictors through a linear combination, and the coefficients of this combination are the regression coefficients (Tümsavaş et al., 2019). Each coefficient is associated to a wavelength and represent its ability to affect the performance of the model to predict the chemical properties of interest (Wang et al., 2015). In addition, the variables that are most important in determining the attributes of interest can be identified through the scores of the variable importance in projection (VIP) (Steidle Neto, Toledo, et al., 2017).

Comparing the regression coefficients and the VIP scores along the wavelength range for Brix and Pol, it reveals visual substantial similarities. Among soluble solids there are inorganic and organic compounds (sugars and non-sugars). Sugars are found in the form of glucose and fructose, called reducing sugars, and sucrose. As already mentioned, in the laboratory, sucrose is measured indirectly by Pol. Brix and Pol are highly correlated attributes (Mancini et al., 2012), probably due to the higher proportion of sucrose in the concentration of soluble solids. This fact explains the similarities found between the response of these attributes.

There are no differences between Brix and Pol for the VIP scores and the regression vector in relation to the wavelengths as expected (Fig. 4), due to the compositional relationship between these attributes. Thus, it is possible to infer the important regions for modelling using these two metrics. The regression vector plots are represented in Fig. 4a for Brix and Fig. 4b for Pol. Similarly, the more sensitive wavelengths to the response variable are represented by the VIP scores for Brix (Fig. 4c) and Pol (Fig. 4d). For VIP values higher than 1, the variables are highly influential (Wold et al., 2001). Absorption bands around 1100 nm and between 1410 and 1440 nm represent the first overtones of OH combinations and stretching, respectively (Golic, Walsh, & Lawson, 2003).

As documented by Golic et al. (2003), in the wavelengths between 1100 nm and 1230 nm the second overtone of CH stretching vibration can occur. It is possible to observe a prominent peak at 1160 nm. Most evident in the middle of the VIP score plots, there is a wide region of absorption bands with greater influence to predict the quality attributes between 1360 nm and 1490 nm, which occurs combinations of vibrational modes involving CH and first overtone of OH and NH chemical bonds (Osborne, 2006). There is a peak with a higher VIP value, compared the wavelengths analysed, at 1390 nm for Pol (Fig. 4d). On the other hand, its value is surpassed by the band at 1730 nm for Brix (Fig. 4c). At 1690 nm and 1730 nm regions, absorbance arising from CH stretching first overtone (Bázár et al., 2015; Golic et al., 2003). In the following bands, between 1720 nm and 1765 nm occurs a peak at 1730 nm. This absorption range is arising from CH_2 stretching first overtone of molecules vibration (Golic et al., 2003).

VIP values lower than 0.8 is insignificant for forecasting (Steidle Neto, Toledo, et al., 2017). There were two absorption intervals in which this occurred with greater intensity: 1210 nm–1340 nm and 1510 nm and 1650 nm. For VIP values between 0.8 and 1.0, there is moderately influential variables. Indeed, when taking into account the analysis of soluble solids content (Brix), there is a low concentration of inorganic, organic non-sugars compounds and reducing sugars (glucose and fructose), besides saccharose content (Pol). The presence of these compounds in the samples is probably the reason for the minimum differences observed between the graphs of these two attributes.

3.3. *Sugarcane quality mapping*

It is evident that NIRS is a suitable tool to provide predictor variables to be used in sugarcane quality forecast models. This fact is corroborated by several studies in the literature (Nawi, Chen, & Jensen, 2013; Phetpan et al., 2018; Phuphaphud et al., 2019; Phuphaphud, Saengprachatanarug, Posom, Maraphum, & Taira, 2020; Taira et al., 2010). In view of this and aware that Brazilian sugarcane farmers are paid based on their quantitative (yield – Mg ha^{-1}) and for the quality of the raw material, it is fairer to farmers to be paid according to the results of each area considering their variability and not being paid by the average value. Thus, sugarcane quality mapping is justified, not only because it provides a more reasonable approach to the buyer and seller, but also, it supports farmers decision-making regarding their crop (Farid, Bakhsh, Ahmad, Ahmad, & Mahmood-Khan, 2016), providing an important layer of information to site-specific management of the crops to meet the precision agriculture objectives (ISPA).

However, a method to obtain high-density data for attributes related to sugarcane quality is not developed, despite the existence of methods to obtain quantitative data (Magalhães & Cerri, 2007; Molin & Menegatti, 2004). Thus, the use of NIRS provides a handy, fast, accurate and reliable method to forecast sugarcane quality attributes. We believe that minimum preparation should solve the issue of variability of contents along the cane stem, as it provides a more uniform and homogeneous sampling. Although, sugarcane harvest is a particular tough environment and the absence of sample prepare is an aspect desirable for an on-board sensor

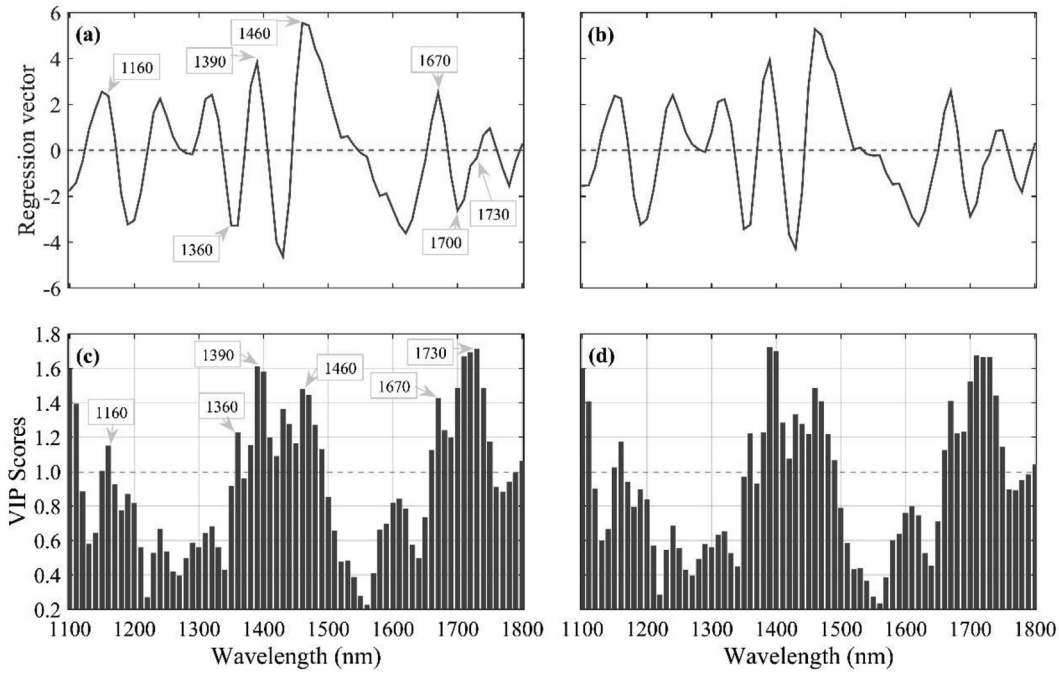

Fig. 4 – Variation of regression coefficients with wavelength, obtained with partial least squares regression (PLSR) analysis of Brix (a) and Pol (b) of sugarcane, and Variable Importance in Projection (VIP) for models used to predict Brix (c) and Pol (d).

development. Some researchers have achieved satisfactory results for solids soluble content prediction in controlled conditions for sugarcane billets measurements (Phetpan et al., 2018; Udompetaikul, Phetpan, & Sirisomboon, 2021). However, it is needed advance for in-field applications.

The maps resulting of laboratory measurements and the maps obtained by NIR measurements show visual spatial similarity for both sugarcane attributes as shown in Fig. 5. These results are expected since the prediction model presented low $RMSE_P$ and high R^2_P for both Pol and Brix content as

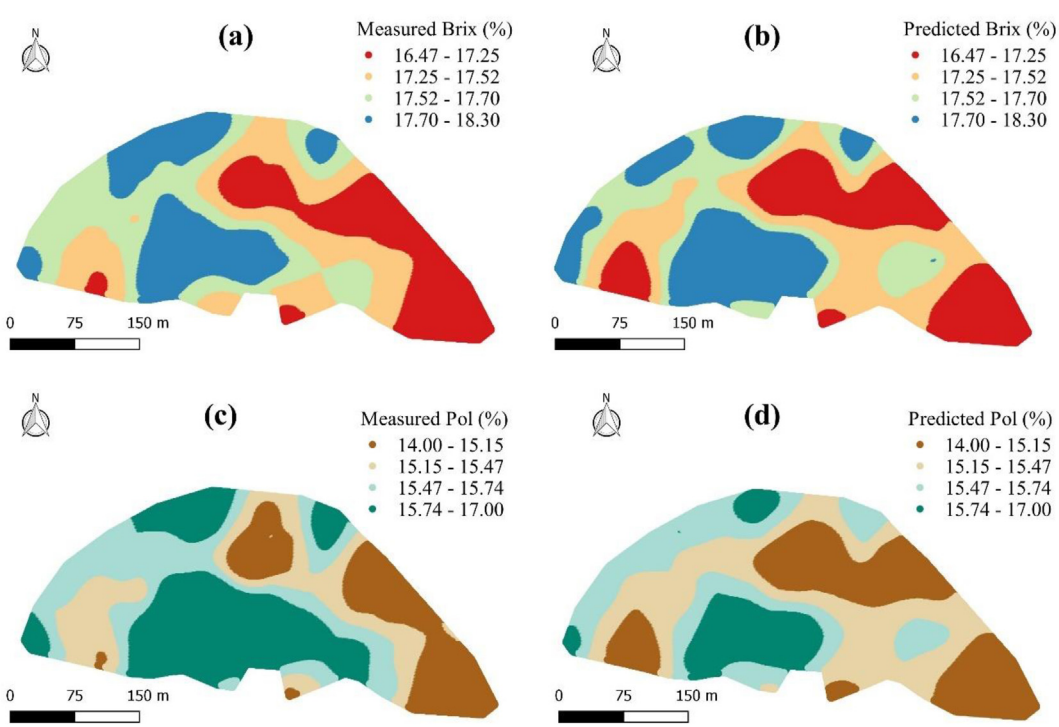

Fig. 5 – Maps of Brix measured (a) and predicted (b), and Pol measured (c) and predicted (d), calculated with partial least squares (PLS) regression.

Table 3 – Variogram parameters for Brix and Pol measured and predicted values.

		Model	C	C_0	$(C + C_0)$	A (m)	SDI (%)
Brix	Measured	Spherical	0.1762	0.2982	0.4744	132.4	62.9
	Predicted	Spherical	0.2511	0.279	0.5301	122.5	52.6
Pol	Measured	Spherical	0.5114	0.4872	0.9986	150.8	48.8
	Predicted	Spherical	0.2723	0.3542	0.6265	124.1	56.5

C: Partial sill, C_0: Nugget, $(C + C_0)$: Sill, A: Range, SDI: spatial dependence index.

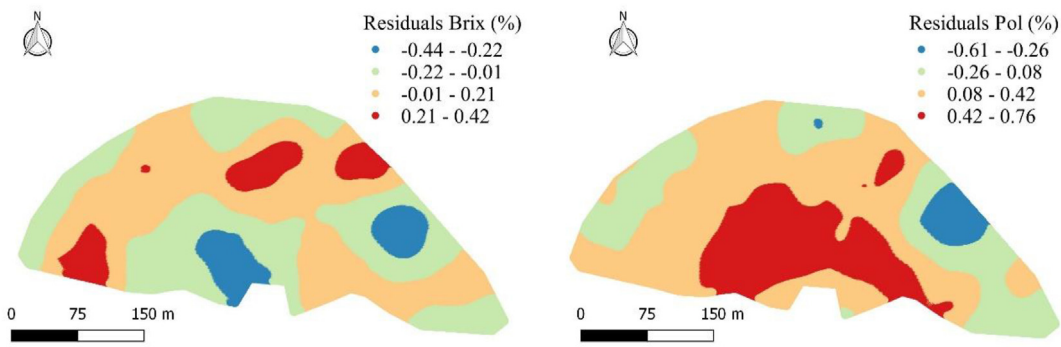

Fig. 6 – Maps of residuals calculated with partial least squares regression (PLSR) for prediction of Brix (a) and Pol (b) of sugarcane. Dark hues show overestimation and light hues show underestimation of the parameter's value.

already seen in Table 2. Initially, there is a region distinguished as having a low potential for Brix and Pol when compared to other regions of the field, extending practically for all the east corner towards the upper central area. On the other hand, there is a region, in the lower central region of the field, that presents high Brix and Pol content in relation to the other areas of the field. Figure 5 highlights the importance for mapping sugarcane quality parameters since there is an uneven distribution for Pol and Brix values along the area. This uneven distribution is expected for both qualitative and quantitative result of the crop as they are the result of a complex interaction among soil properties, weather conditions (macro and microclimate), weed infestations and other factors that present spatial peculiarities (Johnson & Richard, 2005; Sanches et al., 2019).

All geostatistical models presented moderate spatial dependence (Table 3), according to the classification proposed by Cambardella et al. (1994). These results were similar to those obtained by Rodrigues et al. (2013), when analysing the spatial variability of quality attributes using physical and chemical soil attributes and leaf nitrogen as predictors to fit mathematical models to predict Brix, Pol and Fibre. Looking at the range values obtained from predicted and measured models for both Brix and Pol content, the lowest value among all variables was 122.5 m indicating that it is necessary to collect 2.67 samples ha^{-1} to represent faithfully both Brix and Pol content. Thus, comparing the calculated sampling grid (2.67 samples ha^{-1}) to the applied grid (5.50 samples ha^{-1}), it can be concluded that the applied sampling grid was effectively well-designed. In addition to that, in the studied conditions, the results show that there is no need for high density sampling for sugarcane quality analysis indicating that three samples ha^{-1} could be enough. Thus, it opens an opportunity for farmers to adopt sugarcane quality sampling and submit the samples to the NIRS method to predict its qualitative attributes since there is not a developed method to be applied in large scale.

The spatial distribution of the model residuals for Brix and Pol by PLSR are show in Fig. 6. The models tested did not show a similar spatial pattern of residuals across the two spatial parameters. There is a tendency to overestimate (residuals between 0.21 and 0.42%) in areas with low Brix values (Fig. 5a and b, Fig. 6a). On the other hand, for Pol, overestimation in the area with the greatest potential for this attribute is more evident (Fig. 5c and d, Fig. 6b). This overestimation and underestimation are the result of the bias-variance trade-off (Mehta et al., 2019). However, both presented a satisfactory performance with residuals values lower than 0.77%, for under or overestimation. This is easily recognised when looking at the transition areas, ranging from −0.22 to 0.21% for

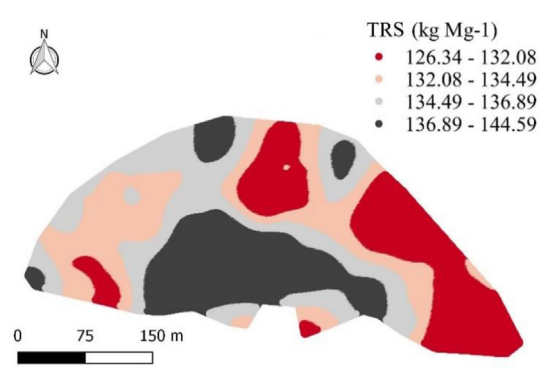

Fig. 7 – Total sugar recoverable map.

Table 4 – Correlation matrix among sugarcane quality maps.

	$Brix_{measured}$	$Brix_{predicted}$	$Pol_{measured}$	$Pol_{predicted}$	$TRS_{measured}$
$Brix_{measured}$	1.00				
$Brix_{predicted}$	0.91	1.00			
$Pol_{measured}$	0.89	0.88	1.00		
$Pol_{predicted}$	0.90	1.00	0.88	1.00	
$TRS_{measured}$	0.89	**0.85**	0.99	**0.85**	1.00

Brix, and −0.26 to 0.42% for Pol, occupying considerably the largest area of the maps.

Given the importance of TRS for sugar producers and sugar mills to price and market the product, and the future perspective of obtaining maps as a tool for managing cause and effect relationships and industrial planning, the possibility of using maps from base attributes (Brix and Pol) to understand the spatial variability of the sugarcane quality was analysed. Figure 7 shows the total recoverable sugar (TRS) map obtained by conventional measurements, in kilograms of sugar recoverable by the industrial process per megagram of sugarcane.

On a first view, there is a visual spatial similarity between TRS (Fig. 7) and measured Pol maps (Fig. 5c). This was expected due to the presence of a high correlation between these attributes (Corrêdo et al., 2021; Ferraz, Corrêdo, Wei, & Molin, 2019). The correlation analysis among TRS, measured and predicted Brix and Pol are shown in Table 4. The correlation between the measured Brix and Pol, and measured TRS were 0.89 and 0.99, respectively. The TRS is obtained by an empirical equation as a function of the pol of cane and reducing sugars of cane, according to CONSECANA (2015). Both attributes are obtained as a function of the pol of juice and reducing sugars of juice, respectively, and as a function inversely proportional to the Fibre content. Despite these relations, in this study, it was not possible to accurately predict Fibre content, therefore, making unfeasible to use this data to predict TRS by NIRS. A viable alternative for future works should concentrate on data processing, evaluating the use of non-linear methods for modelling the Fibre content (Sexton, Everingham, Donald, Staunton, & White, 2018). However, it is clear that TRS presents high correlation with Pol, which is an indirect measure of sucrose content, as mentioned earlier, and presents a low inversely correlation with Fibre content (Mancini et al., 2012).

From the models developed by means of NIRS, the correlation obtained between the interpolated predicted values of Brix and Pol by NIRS and the measured TRS, determined by conventional methods, was 0.85 for both attributes (Table 4, bold numbers). The high correlation presented among attributes, indicate the possibility of applying NIRs technology to measure the quality of sugarcane in the field.

3.4. *Future works*

Future studies with online measurements may assist in advances regarding the application of this technology aiming to obtain high density data. In addition, minimal sample preparation, such as measurements from the stalk, can assist in the implementation of technology (Corrêdo et al., 2021). Some studies pointed to promising results in this direction (Nawi, Chen, & Jensen, 2013; Phetpan et al., 2018; Phuphaphud et al., 2020; Udompetaikul et al., 2021). However, the performance of these systems in field conditions has not yet been evaluated. In addition, the achievement of TRS results by NIRS depends on the development of robust forecast models for Fibre content, in addition to Pol and Brix. Future studies using non-linear methods for modelling multivariate data should help in this regard. Future perspectives on obtaining data related with crop quality will assist in the management of agricultural inputs or even in studies focussing the production of as-apply maps having both layers of information, yield and quality, as a tool to support decision making to evaluate the cause–effect relation between TRS and inputs used in the crop (Corrêdo et al., 2020). Finally, future study with a NIR system embedded in a sugarcane harvester would help to improve the robustness of the technology, enabling the collection of data at high-sampling frequency in sugarcane fields improving the geostatistical analysis to evaluate the spatial variability.

4. Conclusions

NIRS models were developed from parallel with conventional laboratory analyses for determining the quality of sugarcane in a sugar mill. The models developed for Brix and Pol were sufficiently accurate to achieve satisfactory results for the prediction of external validation data, with $RMSE_P$ of 0.80 and 0.58, respectively, and R^2_P of 0.71 for both. However, results obtained for Fibre by NIRS models were not satisfactory (R^2_P of 0.24 and $RMSE_P$ of 1.15).

These data were composed of georeferenced samples obtained in a commercial field of sugarcane to characterise the spatial variability of the attributes through the NIRS. The results obtained in the laboratory and through prediction via NIRS showed spatial dependence. Furthermore, after data interpolation, the maps generated for the attributes from results obtained in the laboratory by conventional analysis obtained a high correlation with the maps generated from the NIRS results (0.91 for Brix and 0.88 for Pol). In addition, both showed a strong correlation with the TRS map generated from the laboratory results (0.85 for both). This study demonstrates that the estimation of sugarcane quality attributes by NIRS allows the analysis of the spatial variability of these attributes in the field. Moreover, future studies should be allowed the requirement for sample preparation be overcome.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors gratefully acknowledge the financial support of São Paulo Research Foundation (Fundação de Amparo à Pesquisa do Estado de São Paulo, FAPESP) for granting a graduate scholarship to the first author (Grant number 2018/25008-8). Also, partially funded by the National Council for Scientific and Technological Development, in the first doctoral scholarship months (Grant number 168645/2017-3). We would like to thank to São Martinho group, for the support provided during data collection.

REFERENCES

ASTM, E1655-17. (2017). *Standard practices for infrared multivariate quantitative analysis* (p. 30). West Conshohocken, Pennsylvania: ASTM International. https://doi.org/10.1520/E1655-17

Bázár, G., Kovacs, Z., Tanaka, M., Furukawa, A., Nagai, A., Osawa, M., et al. (2015). Water revealed as molecular mirror when measuring low concentrations of sugar with near infrared light. *Analytica Chimica Acta, 896*, 52–62. https://doi.org/10.1016/j.aca.2015.09.014

Cambardella, C. A., Moorman, T. B., Novak, J. M., Parkin, T. B., Karlen, D. L., Turco, R. F., et al. (1994). Field-scale variability of soil properties in central Iowa soils. *Soil Science Society of America Journal, 58*(5), 1501–1511. https://doi.org/10.2136/sssaj1994.03615995005800050033x

Colaço, A. F., Molin, J. P., Rosell-Polo, J. R., & Escolà, A. (2019). Spatial variability in commercial orange groves. Part 2: Relating canopy geometry to soil attributes and historical yield. *Precision Agriculture, 20*(4), 805–822. https://doi.org/10.1007/s11119-018-9615-0

CONSECANA. (2015). *Instructions manual* (6th ed., p. 81p). National Concil of Sugarcane Producers of São Paulo State (In Portuguese).

Corrêdo, L. P., Canata, T. F., Maldaner, L. F., de Lima, J. J. A., & Molin, J. P. (2020). Sugarcane harvester for in-field data collection: State of the art, its applicability and future perspectives. *Sugar Tech, 22*(4), 1–14. https://doi.org/10.1007/s12355-020-00874-3

Corrêdo, L.d. P., Maldaner, L. F., Bazame, H. C., & Molin, J. P. (2021). Evaluation of minimum preparation sampling strategies for sugarcane quality prediction by vis-NIR spectroscopy. *Sensors, 21*(6), 2195. https://doi.org/10.3390/s21062195

Cortés, V., Blasco, J., Aleixos, N., Cubero, S., & Talens, P. (2019). Monitoring strategies for quality control of agricultural products using visible and near-infrared spectroscopy: A review. *Trends in Food Science & Technology, 85*, 138–148. https://doi.org/10.1016/j.tifs.2019.01.015

Farid, H. U., Bakhsh, A., Ahmad, N., Ahmad, A., & Mahmood-Khan, Z. (2016). Delineating site-specific management zones for precision agriculture. *Journal of Agricultural Science, 154*(2), 273–286. https://doi.org/10.1017/S0021859615000143

Ferraz, M. N., Corrêdo, L. P., Wei, M. C. F., & Molin, J. P. (2019). Spatial variability mapping of sugarcane qualitative attributes. *Engenharia Agrícola, 39*(spe), 109–117. https://doi.org/10.1590/1809-4430-eng.agric.v39nep109-117/2019

Golic, M., Walsh, K., & Lawson, P. (2003). Short-wavelength near-infrared spectra of sucrose, glucose, and fructose with respect to sugar concentration and temperature. *Applied Spectroscopy, 57*(2), 139–145. https://doi.org/10.1366/000370203321535033

International commission for uniform methods of sugar analysis (ICUMSA). Available online: https://www.icumsa.org. (Accessed 1 December 2020).

International society of precision agriculture (ISPA). Available online: https://www.ispag.org. (Accessed 1 April 2020).

Johnson, R. M., & Richard, E. P. (2005). Sugarcane yield, sugarcane quality, and soil variability in Louisiana. *Agronomy Journal, 97*(3), 760–771. https://doi.org/10.2134/agronj2004.0184

Magalhães, P. S. G. G., & Cerri, D. G. P. P. (2007). Yield monitoring of sugar cane. *Biosystems Engineering, 96*(1), 1–6. https://doi.org/10.1016/j.biosystemseng.2006.10.002

Maldaner, L. F., & Molin, J. P. (2020). Data processing within rows for sugarcane yield mapping. *Scientia Agricola, 77*(5), 2020. https://doi.org/10.1590/1678-992x-2018-0391

Mancini, M. C., Leite, D. C., Perecin, D., Bidóia, M. A. P., Xavier, M. A., Landell, M. G. A., et al. (2012). Characterization of the genetic variability of a sugarcane commercial cross through yield components and quality parameters. *Sugar Technology, 14*(2), 119–125. https://doi.org/10.1007/s12355-012-0141-5

Maraphum, K., Chuan-Udom, S., Saengprachatanarug, K., Wongpichet, S., Posom, J., Phuphaphud, A., et al. (2018). Effect of waxy material and measurement position of a sugarcane stalk on the rapid determination of Pol value using a portable near infrared instrument. *Journal of Near Infrared Spectroscopy, 26*(5), 287–296. https://doi.org/10.1177/0967033518795810

Mehmood, T., & Ahmed, B. (2016). The diversity in the applications of partial least squares: An overview. *Journal of Chemometrics, 30*(1), 4–17. https://doi.org/10.1002/cem.2762

Mehta, P., Bukov, M., Wang, C. H., Day, A. G. R., Richardson, C., Fisher, C. K., et al. (2019). A high-bias, low-variance introduction to Machine Learning for physicists. *Physics Reports, 810*, 1–124. https://doi.org/10.1016/j.physrep.2019.03.001

Minasny, B., Mcbratney, A. B., & Whelan, B. M. (2005–2006). VESPER 1.5 - spatial prediction software for precision agriculture. In *Australian centre for precision agriculture, McMillan building A05*. NSW: The University of Sydney. http://sydney.edu.au/agriculture/pal/software/vesper.shtml.

Molin, J. P., & Menegatti, L. A. A. (2004). Field-testing of a sugar cane yield monitor in Brazil. *ASAE Annual International Meeting 2004, 300*(4), 733–744. https://doi.org/10.13031/2013.16159

Morellos, A., Pantazi, X. E., Moshou, D., Alexandridis, T., Whetton, R., Tziotzios, G., et al. (2016). Machine learning based prediction of soil total nitrogen, organic carbon and moisture content by using VIS-NIR spectroscopy. *Biosystems Engineering, 152*, 104–116. https://doi.org/10.1016/j.biosystemseng.2016.04.018

Nawi, N. M., Chen, G., & Jensen, T. (2013). Visible and shortwave near infrared spectroscopy for predicting sugar content of sugarcane based on a cross-sectional scanning method. *Journal of Near Infrared Spectroscopy, 21*(4), 289–297. https://doi.org/10.1255/jnirs.1060

Nawi, N. M., Chen, G., & Jensen, T. (2014). In-field measurement and sampling technologies for monitoring quality in the sugarcane industry: A review. *Precision Agriculture, 15*(6), 684–703. https://doi.org/10.1007/s11119-014-9362-9

Nawi, N. M., Chen, G., Jensen, T., & Mehdizadeh, S. A. (2013). Prediction and classification of sugar content of sugarcane based on skin scanning using visible and shortwave near

infrared. *Biosystems Engineering, 115*, 154–161. https://doi.org/10.1016/j.biosystemseng.2013.03.005

Oliveri, P., Malegori, C., Simonetti, R., & Casale, M. (2019). The impact of signal pre-processing on the final interpretation of analytical outcomes – a tutorial. *Analytica Chimica Acta, 1058*, 9–17. https://doi.org/10.1016/j.aca.2018.10.055

Osborne, B. G. (2006). Near-infrared spectroscopy in food analysis. *Encyclopedia of Analytical Chemistry, 14*. https://doi.org/10.1002/9780470027318.a1018

Pasquini, C. (2018). Near infrared spectroscopy: A mature analytical technique with new perspectives – a review. *Analytica Chimica Acta, 1026*, 8–36. https://doi.org/10.1016/j.aca.2018.04.004

Phetpan, K., Udompetaikul, V., & Sirisomboon, P. (2018). An online visible and near-infrared spectroscopic technique for the real-time evaluation of the soluble solids content of sugarcane billets on an elevator conveyor. *Computers and Electronics in Agriculture, 154*, 460–466. https://doi.org/10.1016/j.compag.2018.09.033

Phuphaphud, A., Saengprachatanarug, K., Posom, J., Maraphum, K., & Taira, E. (2019). Prediction of the fibre content of sugarcane stalk by direct scanning using visible-shortwave near infrared spectroscopy. *Vibrational Spectroscopy, 101*, 71–80. https://doi.org/10.1016/j.vibspec.2019.02.005

Phuphaphud, A., Saengprachatanarug, K., Posom, J., Maraphum, K., & Taira, E. (2020). Non-destructive and rapid measurement of sugar content in growing cane stalks for breeding programmes using visible-near infrared spectroscopy. *Biosystems Engineering, 197*, 76–90. https://doi.org/10.1016/j.biosystemseng.2020.06.012

RIDESA. (2015). Liberação nacional de variedades RB de cana-de-açúcar. *Rede Interuniversitária para Desenvolvimento do Setor Sucroalcooleiro, 77* (In Portuguese).

Rodrigues, F. A., Magalhães, P. S. G., & Franco, H. C. J. (2013). Soil attributes and leaf nitrogen estimating sugar cane quality parameters: Brix, pol and fibre. *Precision Agriculture, 14*(3), 270–289. https://doi.org/10.1007/s11119-012-9294-1

Sanches, G. M., Magalhães, P. S. G., & Franco, H. C. J. (2019). Site-specific assessment of spatial and temporal variability of sugarcane yield related to soil attributes. *Geoderma, 334*, 90–98. https://doi.org/10.1016/j.geoderma.2018.07.051

Sanseechan, P., Panduangnate, L., Saengprachatanarug, K., Wongpichet, S., Taira, E., & Posom, J. (2018). A portable near infrared spectrometer as a non-destructive tool for rapid screening of solid density stalk in a sugarcane breeding program. *Sensing and Bio-Sensing Research, 20*, 34–40. https://doi.org/10.1016/j.sbsr.2018.07.001

Savitzky, A., & Golay, M. J. E. (1964). Smoothing and differentiation of data by simplified least squares procedures. *Analytical Chemistry, 36*(8), 1627–1639. https://doi.org/10.1021/ac60214a047

Sexton, J., Everingham, Y., Donald, D., Staunton, S., & White, R. (2018). A comparison of non-linear regression methods for improved on-line near infrared spectroscopic analysis of a sugarcane quality measure. *Journal of Near Infrared Spectroscopy*. https://doi.org/10.1177/0967033518802448

Sorol, N., Arancibia, E., Bortolato, S. A., & Olivieri, A. C. (2010). Visible/near infrared-partial least-squares analysis of Brix in sugar cane juice. A test field for variable selection methods. *Chemometrics and Intelligent Laboratory Systems, 102*(2), 100–109. https://doi.org/10.1016/j.chemolab.2010.04.009

Steidle Neto, A. J., Lopes, D. C., Pinto, F. A. C., & Zolnier, S. (2017). Vis/NIR spectroscopy and chemometrics for non-destructive estimation of water and chlorophyll status in sunflower leaves. *Biosystems Engineering, 155*, 124–133. https://doi.org/10.1016/j.biosystemseng.2016.12.008

Steidle Neto, A. J., Toledo, J. V., Zolnier, S., de Lopes, D. C., Pires, C. V., & Silva, T. G. F.d. (2017). Prediction of mineral contents in sugarcane cultivated under saline conditions based on stalk scanning by Vis/NIR spectral reflectance. *Biosystems Engineering, 156*, 17–26. https://doi.org/10.1016/j.biosystemseng.2017.01.003

Taira, E., Ueno, M., & Kawamitsu, Y. (2010). Automated quality evaluation system for net and gross sugarcane samples using near infrared spectroscopy. *Journal of Near Infrared Spectroscopy, 18*(3), 209–215. https://doi.org/10.1255/jnirs.884

Tümsavaş, Z., Tekin, Y., Ulusoy, Y., & Mouazen, A. M. (2019). Prediction and mapping of soil clay and sand contents using visible and near-infrared spectroscopy. *Biosystems Engineering, 177*, 90–100. https://doi.org/10.1016/j.biosystemseng.2018.06.008

Udompetaikul, V., Phetpan, K., & Sirisomboon, P. (2021). Development of the partial least-squares model to determine the soluble solids content of sugarcane billets on an elevator conveyor. *Measurement: Journal of the International Measurement Confederation, 167*, 107898. https://doi.org/10.1016/j.measurement.2020.107898

Valderrama, P., Braga, J. W. B., & Poppi, R. J. (2007). Validation of multivariate calibration models in the determination of sugar cane quality parameters by near infrared spectroscopy. *Journal of the Brazilian Chemical Society, 18*(2), 259–266. https://doi.org/10.1590/S0103-50532007000200003

Wang, J., Nayak, S., Koch, K., & Ming, R. (2013). Carbon partitioning in sugarcane (Saccharum species). *Frontiers of Plant Science, 4*, 201. https://doi.org/10.3389/fpls.2013.00201

Wang, H., Peng, J., Xie, C., Bao, Y., & He, Y. (2015). Fruit quality evaluation using spectroscopy technology: A review. *Sensors, 15*(5), 11889–11927. https://doi.org/10.3390/s150511889

Wold, S., Sjöström, M., & Eriksson, L. (2001). PLS-regression: A basic tool of chemometrics. *Chemometrics and Intelligent Laboratory Systems, 58*(2), 109–130. https://doi.org/10.1016/S0169-7439(01)00155-1